

INT422 IT Project Management

Unit II

Project Management Features - Risk Analysis, Project Control, Project Audit and Project Termination.

Lecture 15

Outline

Risk Analysis

Project Control

Project Audit and Project

Termination

Risk Analysis

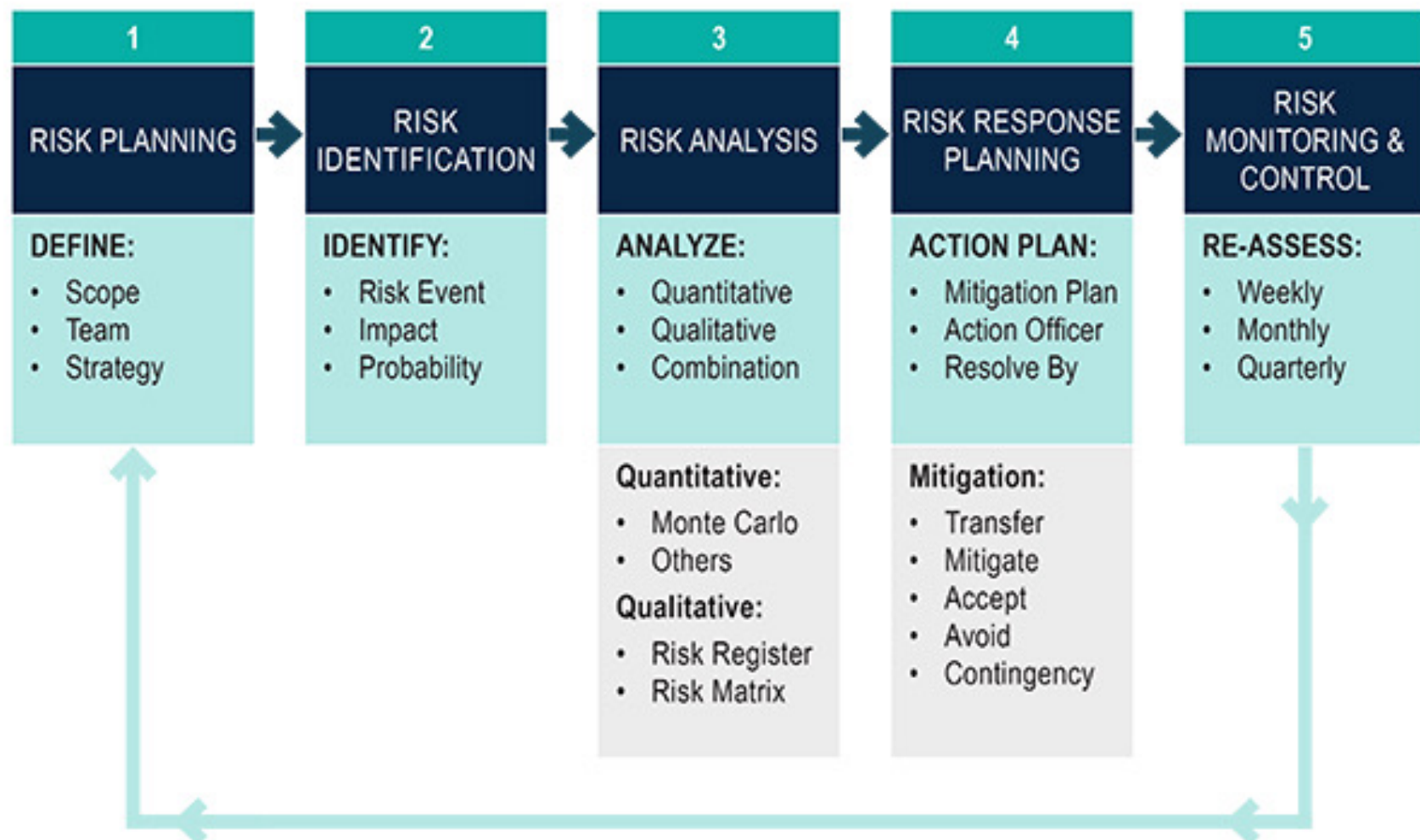
Risk analysis is about developing an understanding of the risk. Some of the major risk factors for Port projects are:

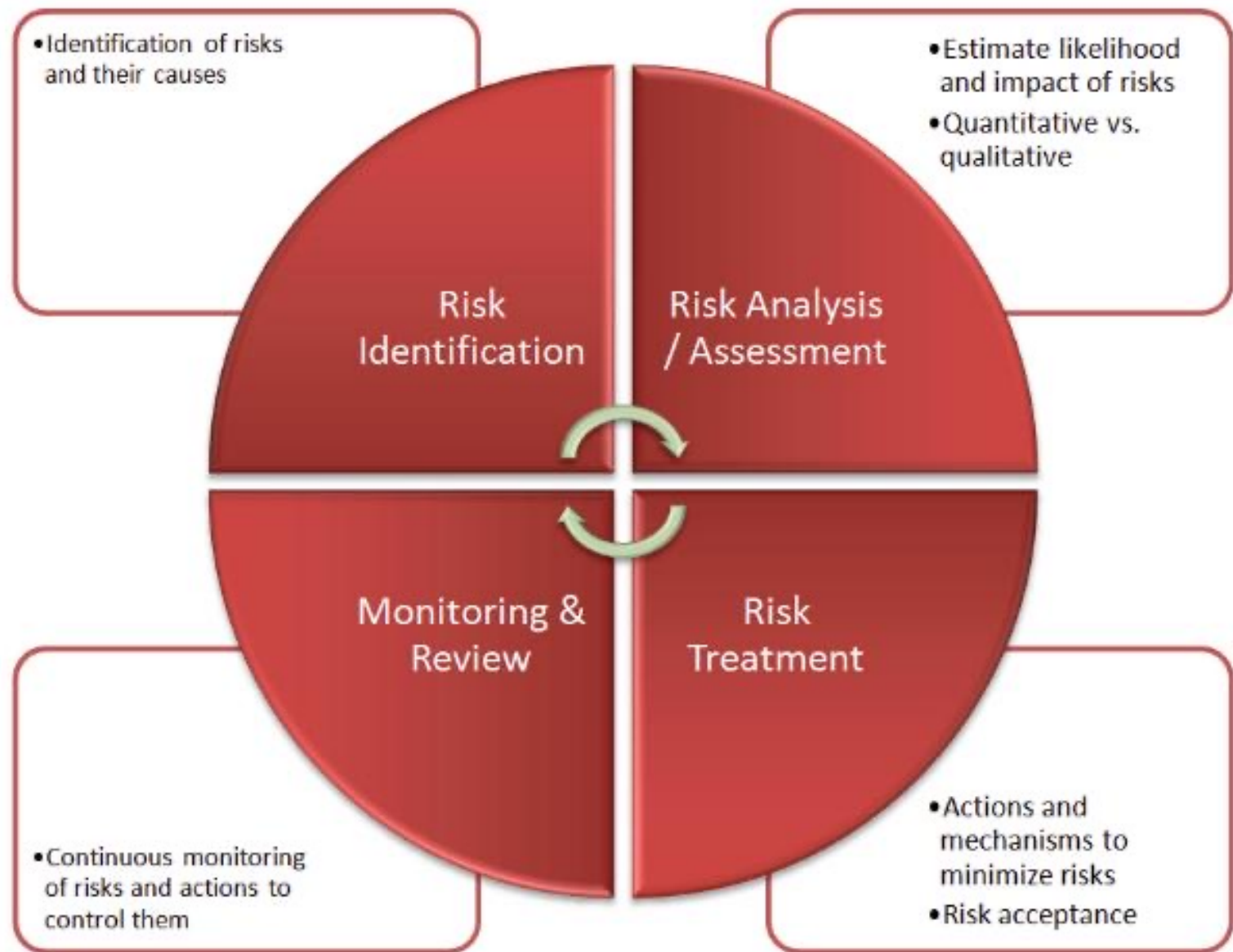
- They require major political involvement
- The project budget is overrun
- The project completion time is overrun
- There is a failure to achieve anticipated benefits
- Cruise ship boycott
- Natural disasters

Risk Analysis

The main risk factors that are focused on in options analysis are consequence and likelihood.

Consequence x likelihood = risk rating





Risk Analysis

Likelihood	Consequences				
	Insignificant	Minor	Moderate	Major	Severe
Almost Certain	Medium	High	High	Extreme	Extreme
Likely	Medium	Medium	High	High	Extreme
Possible	Low	Medium	Medium	High	Extreme
Unlikely	Low	Medium	Medium	Medium	High
Rare	Low	Low	Medium	Medium	High

Options Analysis Summary

60%

Normal

H13

A B C D E F G H I J K L M N O P Q R S T U



Internal Economic	25%
External Economic	25%
Value Score	25%
Qualitative	25%
Total	100%

Raw Input

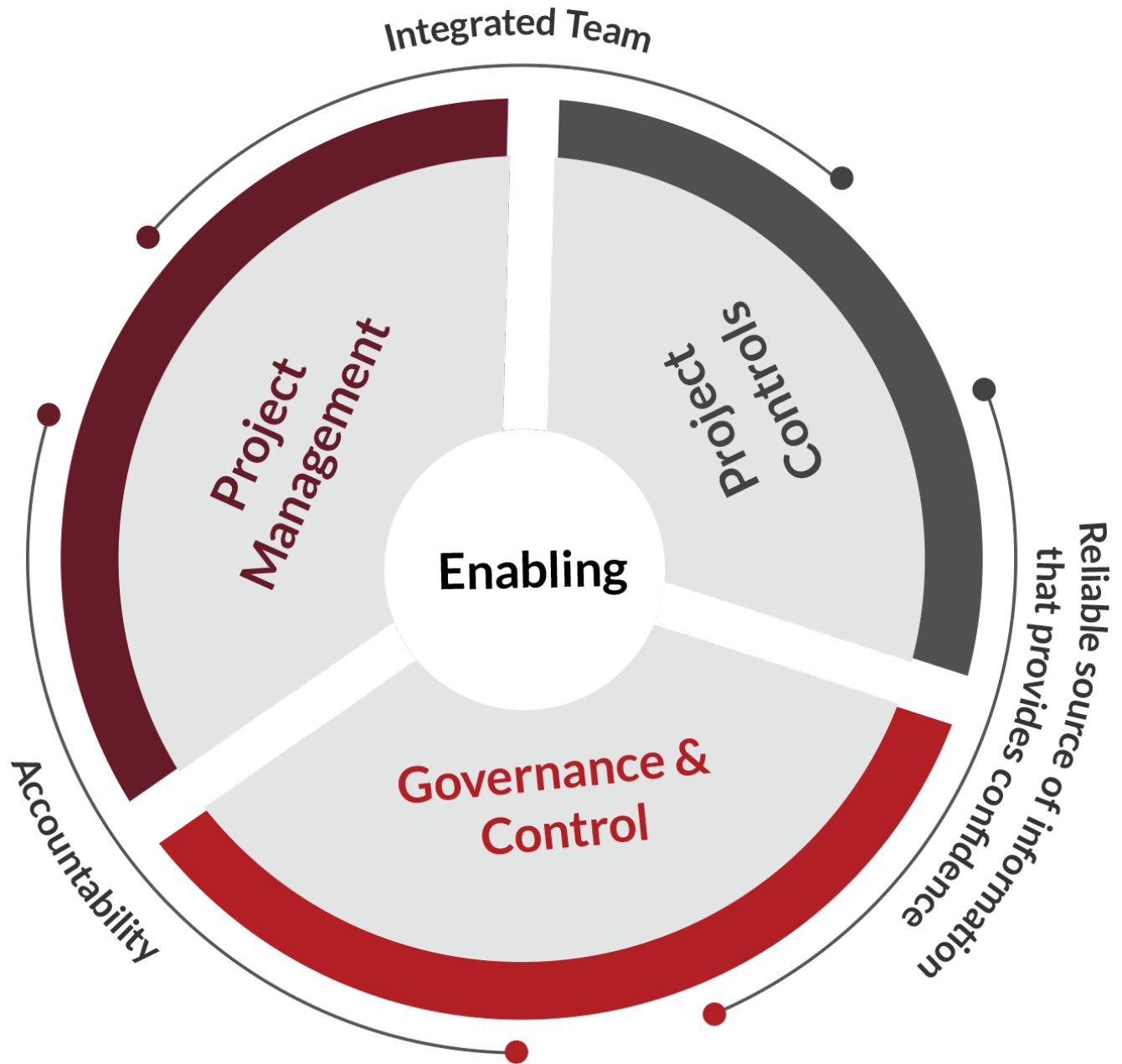
		Internal Economic			External Economic	Value
		NPV			Score	Value Score
		Low	Most Likely	High		
Option 1	Bank Funding	n/a	n/a	n/a	n/a	n/a
Option 2	Public Private Partnership	n/a	n/a	n/a	n/a	n/a
Option 3	Sinking Fund	n/a	n/a	n/a	n/a	n/a
Option 4	Bond	n/a	n/a	n/a	n/a	n/a

Weighted Final Scores

		Internal Economic			External Economic	Value	Total	Rank
		NPV			Score	Value Score	Total Score	Rank Score
		Low	Most Likely	High				
Option 1	Bank Funding	0.00	0.00	0.00	0.00	0	0.00	1
Option 2	Public Private Partnership	0.00	0.00	0.00	0.00	0	0.00	1
Option 3	Sinking Fund	0.00	0.00	0.00	0.00	0	0.00	1
Option 4	Bond	0.00	0.00	0.00	0.00	0	0.00	1

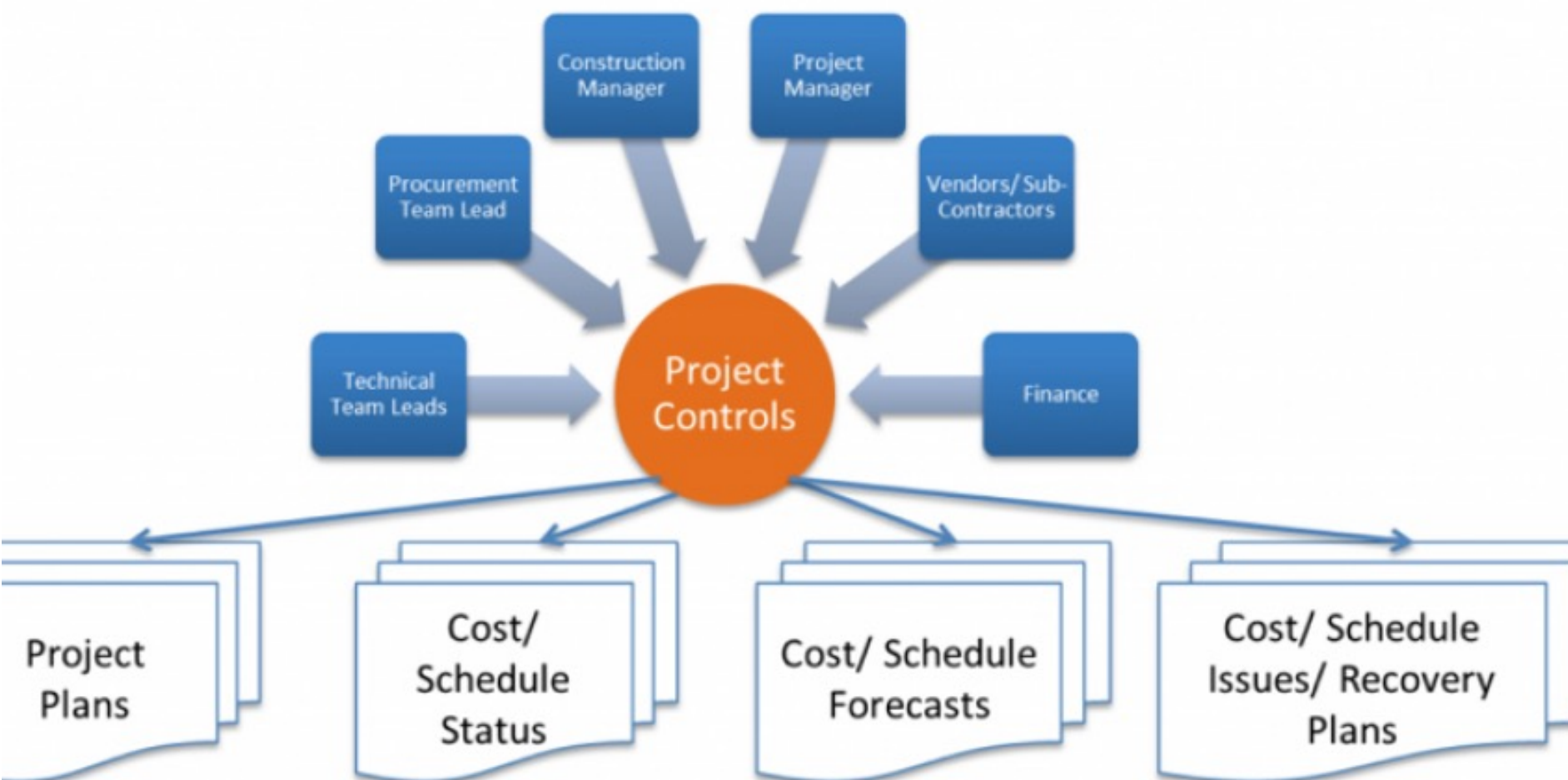
Project Control

- According to the [Project Management Body of Knowledge \(PMBOK\)](#), project control consists of data collection management and analysis processes used to predict and constructively influence the time and cost results of a (large scale) project or programme by communicating information in formats that support effective management and decision-making.
- It goes without saying that Project Control looks different depending on the project and the management plan.
- It is a professional function that is not generally recognized as part of the whole of a specialized skill.
- Above all, it is a function that is essential for realizing and delivering project results and benefits in terms of cost, performance, and time.



Project Control

- Although this method can be applied to many more elements of project management, the following list represents the main areas of Project Control:
- [Planning](#)
- [Risk management](#)
- Cost management
- [Scope](#) and change management
- [Earned Value Management](#)
- [Performance management](#)
- [Stakeholder management](#)
- Reporting control



Why is Project Control important?

- Project management and PC have both existed since the mid-twentieth century. Throughout the years, it has been demonstrated time and again how poor project planning and poor project monitoring almost inevitably results in project failure.
- Management of the project, if done properly, saves both time and money and leads to a successful project. Project Control should therefore be a leading function of every project.
- Before this article goes deeper into the method, we will first explain the general terms regarding project management.

Project Control in the planning phase

- The initial stage of a project is mainly about project planning / project scope. Here, the basic project management principles of a project are outlined, what problems need to be solved, the people (team members) who need to be involved in the project, and what tasks will actually be performed.
 - The different tasks associated with a project
 - The duration of each project phase
 - The budgets for all individual tasks
 - How the activities are distributed among the team members

Project Control in the development phase

- The development phase of project management is mainly about getting the ball rolling. During this phase, the project team meets for the first time, together with stakeholders, and the assignments and activities are planned. In this phase too, PC is key to success. Cost estimates and timetables are converted into realistic planning, and the most important project controls are established here. These are:
 - Planning
 - Cost management
 - Cost forecasts
 - Project Risk Management

Project Control in the implementation phase

- Many projects use a team management tool to quickly identify time drains, quality issues, and other problems. In any case, make sure that the following matters are analyzed:
 - Individual and total employee capacity
 - Distribution of activities per employee / project stakeholders
 - How much time employees spend on each task
 - Number of hours worked per employee / project stakeholders
 - Expenses per employee / project stakeholders

Project Control in the closing phase

- In the closing phase of a project, the project is formally concluded, and the sponsor is informed of the overall level of success. Moreover, in this phase the products or services to be supplied are transferred to the customer.
- The fact that the project is complete does not mean that no more project checks have to be performed. To execute projects more effectively in the future, the final performance of the project team must be assessed.
- Which trends and processes have led to success? What should have been different? What could have been more efficient? Has the team achieved the goals? Have they kept the expected pace? Is the end result within the limits of quality and costs?

Advantages of Project Control

- Provides increased predictability for costs and deadlines.
- Provides more and better insights into the financial health of a project in all of its phases.
- Allows one to more easily prevent postponement of results.
- Ensures higher margins.
- Ensures an improved reputation when it comes to project management and result guarantees.
- Provides an increased sense of employee satisfaction.
- Provides a competitive advantage over project teams with less mature project management capabilities.

Project Audit

- The term *audit* is synonymous with analysis, checkup, inspection, review, scan, scrutiny, survey, and view. Audits are used throughout the project life cycle as follows:
- Project audit
- Scheduled audits
- Unscheduled audits
- Subcontractor and vendor audits
- Quality audits
- Project safety audits

Auditing process

Planning



Design



Fieldwork



Reporting &
recommendations



Corrective actions

Project Audit

- The project audit thoroughly examines:
- Project management
- Methodologies and procedures
- Records
- Properties
- Budgets and expenditures
- Degree of completion

Project Audit

- A formal review of any aspect of a project
- Audits focus on whatever matters senior management desires—Another type of audit is an ethics audit
- Evaluate means to set the value of or appraise
- Project evaluation appraises progress and performance against standard

Purpose of Evaluation—Goals of the System

- Efficiency in meeting both the budget and the schedule
- Customer impact/satisfaction
- Business/direct success
- Future potential
- The main purpose of an audit is to help achieve the goals of the project
- All facets of the project are studied
- A project audit is equivalent to the application of TQM to project management

Approach to Project Audit

- All facets of the project are studied
- The strengths and weaknesses are identified
- Recommendations are prepared to help current and future projects

Project Audit Recommendations

- Identify problems earlier
- Clarify scope, cost, and time relationships
- Improve performance
- Locate technological advances
- Evaluate quality
- Reduce costs
- Improve risk identification
- Many more...

Direct and Ancillary Project Objectives

- Direct goals are stated project objectives—
- Including customer satisfaction
- Direct goals ignore many costs and benefits to:—
 - The project—Its team members—The parent organization
- Unstated objectives are called ancillary goals

Depth of the Audit

- Time and money limit the depth of an audit
- Audits are distracting to those working on the project
- A poor audit result will lower morale on the project

Timing of the Audit

- All significant projects should be audited
- Larger projects may be audited several times
- An audit may also be conducted after the project is over (post-project audits)

Format and Use of the Audit Report

- It should facilitate the comparison of actual versus predicted results
- Significant deviations should be highlighted
- Reasons for significant deviations should be given
- Plans for resolving negative deviations should be discussed

Responsibilities of the Project Auditor

- Be honest and ethical
- Be independent
- Tell the whole truth
- Seek help for technical issues

The Project Audit Life Cycle

- Project audit initiation
- Project baseline definition
- Establishing an audit database
- Preliminary analysis of the project
- Audit report preparation
- Project audit termination



Project Termination

- All projects end—
- The objectives have been completed—It no longer makes sense to finish
- Some teams move on to other projects
- Other times, members go their own way
- The client may be happy, mad, or anywhere in between

Varieties of Project Termination•

- Termination by extinction
- Termination by addition
- Termination by integration
- Termination by starvation

The Varieties of Project Termination

- Termination by extinction
- Termination by addition
- Termination by integration
- Termination by starvation

Termination by Extinction

- Extinction occurs in any scenario where the project goes away
 - Successful
 - Unsuccessful
 - Changes in environment
 - Take too long
 - Murder
- When work on a project stops, some organizational work continues

Termination by Addition

- Applies to an in house project
- When the project is successful, it is institutionalized
- While the project goes away, project personnel and assets are transferred to the new business

Termination by Integration

- The most common way to terminate a project
- The project comes into the business
- It is absorbed into the existing structure
- That structure absorbs the assets of the project

Aspects of the Transition from Project to Integrated Operation

- Personnel
- Manufacturing
- Accounting/finance
- Engineering
- Information systems
- Marketing
- Purchasing
- Risk management

Termination by Starvation

- Termination by starvation involves greatly reducing the budget of a project
- Used when it is politically dangerous to cancel a project
- Bad manners to enquire the status of the project

When to Terminate a Project

- Projects take on a life of their own
- It may be easy to terminate a project that is finished
- But it can be very difficult to terminate a project prior to its completion

Critical Success Factors

- Project mission
- Top-management support
- Project schedule/plan
- Client consolation
- Personnel
- Technical tasks
- Client acceptance
- Monitoring and feedback
- Communication
- Trouble-shooting

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Non-Technical Reasons for Termination

- Political
- Cross
- Cultural
- Senescence

The Final Report—A Project History

- Project performance
- Administrative performance
- Organizational structure
- Project and administrative teams
- Techniques of project management